ANALYTICAL CUMULATIVE EXAM

December 4, 1993

As functioning parts of an instrument, each component contributes some error to the measurement of the property of interest. Viewed as a dynamic system, a measuring device can be no more reliable than its least accurate part. Consideration of the origin and propagation of error becomes acute in the design, modification, or repair of instruments. Many instruments present an output that was calculated from several components, in this exam you will be asked to determine the uncertainty in that output.

Consider an instrument which outputs a value, G, which is a function of variables A,B,C,D or G(A,B,C,D)

a. (20 points) Write a general expression for the total differential dG of G.

b. (10 points) Write the expression for $(dG)^2$.

c. (20 points) In general, the uncertainty or scatter in a measured value x, is given by.

$$\sigma_x^2 = \frac{\Sigma\,(dx)^2}{n}$$

transform the equation in part b from one of differentials, to one of σ^2 noting that all of the variables behave independently of each other and positive and negative values of dA, dB, dC, dD are all equally probable.

d. (20 points) Consider an instrument which calculates the value of G, given by the relationship:

$$G = \frac{AB^2\,(C + 1)}{D^3}$$

in which A, B, C, and D are parameters which are internally generated in the instrument. Derive the specific equation for σ_G^2 based on your general expression derived in part c.

e. (10 points) The values for A,B,C,D are listed be low with their associated sigma values.

A = 20.0 ± 0.1
B = 5.00 ± 0.05
C = 25.0 ± 0.1
D = 10.0 ± 1.0

Calculate σ_G, the absolute uncertainty in G.

f. (10 points) Calculate σ_G/G, the relative uncertainty in G.

g. (10 points) Indicate the order in which you would attack the four variable parameters if you needed to improve the precision of G. Explain your logic.

Helpful Relationships:

$$du^n = nu^{n-1}du$$

$$duv = udv + vdu$$

$$d\ (\ln u) = \frac{1}{u}du$$

$$d\left(\frac{u}{v}\right) = \frac{vdu - udv}{v^2}$$